

Final Project

Introduction

Dementia is the seventh leading cause of death, affecting over 55 million people worldwide [1]. The onset of dementia significantly impacts not only the affected individuals but also their families and healthcare systems, making it an urgent area for research and intervention. Traumatic brain injury (TBI) is a major risk factor for dementia, especially Alzheimer's disease. Epidemiological studies have shown that individuals with a history of TBI, particularly those who experience repeated concussions, have an increased risk of developing dementia and related neurodegenerative conditions [2].

The Aging, Dementia, and Traumatic Brain Injury (TBI) study [3] focuses on understanding the relationship between TBI, neuropathological markers, and dementia in an aged population-based cohort. The dataset from this study includes de-identified clinical data from a cohort of individuals, including various neuropathological measures (Braak stage, CERAD score, ApoE4 allele presence) and clinical diagnoses such as Alzheimer's disease and other dementias. In this paper, the following question is investigated: Can we predict the diagnosis of dementia based on clinical and neuropathological variables, and are there distinct subgroups within the population based on these variables? The analyses will implement machine learning algorithms to predict dementia status and uncover potential hidden patterns in the data, which may inform future research and clinical practice.

Methods

The dataset includes key variables including age, years of education, ApoE4 allele status, TBI history, Braak stage, and CERAD score. Age was a categorical variable with levels such as "77," "89," and "90-94." To conduct the analysis, age was converted to a numerical variable and for any given ranges, the midpoint was utilized. For example, observations with the level "90-94" were recoded to be 92. Additionally, dementia status was refactored so that the "No Dementia" and "Dementia" levels were internally coded as 1 and 2, respectively, in R to reduce potential coding errors. Any missing values in the data were either imputed or removed based on their relevance to the analysis. After preprocessing, the dataset reduced from 107 observations to 100 with 20 variables. The data was split into training (70% = 70 observations) and test (30% = 30 observations) sets. I used logistic regression and random forest classifiers to predict dementia based on clinical features including age, years of education, ApoE4 allele status, number of previous TBIs with loss of consciousness, Braak stage, and CERAD score. The logistic regression model was fit to the training data using the `glm` function in R. The model predicts the binary outcome of dementia (No Dementia or Dementia) using the specified predictors. Random forest was also implemented to potentially improve upon the logistics regression model's performance and handle overfitting. The random forest model was trained with 500 trees (`ntree` = 500) and an optimal number of features tried at each split (`mtry` = 3). Feature importance was calculated to identify the most influential variables. The `randomForest` function was used to fit the model and assess performance.

Both models were evaluated using a test set, and performance metrics such as accuracy, sensitivity, specificity, and the confusion matrix were computed to assess their predictive capabilities.

A k-means clustering algorithm was applied with the goal of identifying potential subgroups at varying risk for dementia. The number of clusters was determined by evaluating the within-cluster sum of squares (WSS) and silhouette scores. The resulting clusters were analyzed for associations with ApoE4 allele status and other clinical variables.

Results

The table presents the odds ratios and its corresponding 95% confidence intervals and p-values for each predictor in the logistic regression model on the training set. This revealed that the presence of the ApoE4

allele had an odds ratio of 2.68, meaning people with the ApoE4 allele have 2.7 times the odds of dementia compared to those without, though this is not statistically significant (p-value = 0.1371).

Table 1 - Odds Ratios of Predictors of Dementia Status

<i>Predictors</i>	act_demented		
	<i>Odds Ratios</i>	<i>CI</i>	<i>p</i>
(Intercept)	3.62	0.00 – 41494.21	0.780
age numeric	0.98	0.89 – 1.07	0.684
education years	0.91	0.76 – 1.08	0.287
apo e4 allele [Y]	2.68	0.75 – 10.54	0.137
num tbi w loc	1.02	0.42 – 2.38	0.973
braak	1.46	0.99 – 2.23	0.063
cerad	1.02	0.57 – 1.84	0.938
Observations	70		
R ² Tjur	0.143		

The logistic regression model on the test data had an accuracy of 66.67% (95% CI: 47.19%–82.71%) with a p-value of 0.04937. The model has a sensitivity of 73.33% and specificity of 60%. The McNemar's test p-value of 0.75183 indicates that there is no significant difference between the errors made for predicting dementia versus no dementia. The results suggest that while the model performs reasonably well, there is still room for improvement, particularly in specificity and overall agreement. The AUC value was 0.724, indicating that the model does an acceptable job at distinguishing between dementia and no dementia.

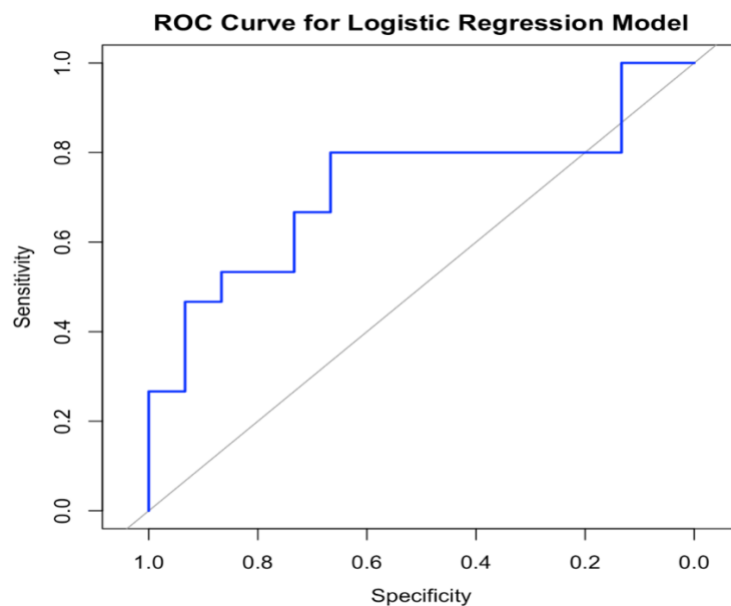


Figure 1.

In comparison, the random forest model, with 500 trees and `mtry` = 3, performed far better with an accuracy of 76.67% (95% CI: 57.72%–90.07%), which was significantly greater than the no-information rate (p-value = 0.002611). The confusion matrix showed that the random forest model had a sensitivity of 73.33% and a specificity of 80%. The positive predictive value (PPV) was 78.57%, and the negative predictive value (NPV) was 75%. The Kappa statistic was 0.53, suggesting moderate to substantial agreement. Feature importance analysis also revealed that Braak stage was the most important predictor of dementia, followed by education years, age, and ApoE4 allele presence. The AUC score was higher than that of the logistic regression model with a value of 0.8311. This means the random forest model does better at classifying between dementia cases.

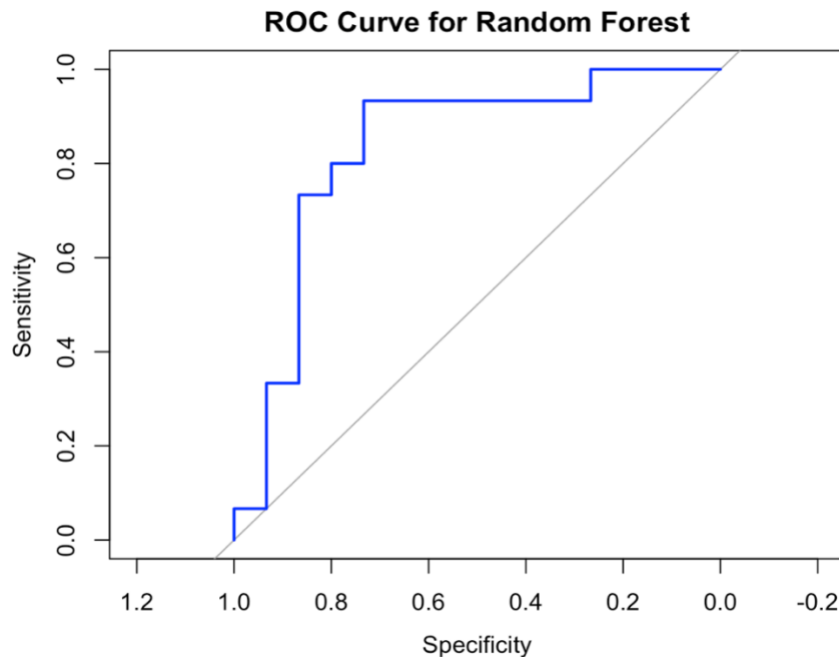


Figure 2.

K-means clustering was performed to identify natural subgroups within the data. The optimal number of clusters ($k = 2$) was determined using the within-cluster sum of squares (WSS) and silhouette score.

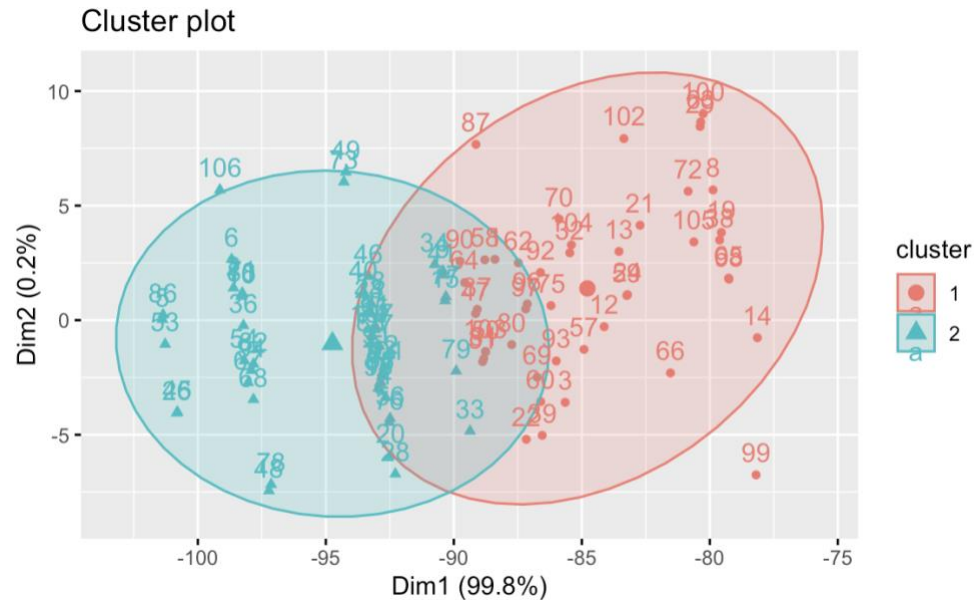


Figure 3.

The resulting clusters were compared with the presence of the ApoE4 allele. The first cluster comprised of 35 individuals (76%) with "No" ApoE4 allele and 11 individuals (24%) with the "Yes" ApoE4 allele. Cluster 2 consisted of 45 individuals (83%) with "No" ApoE4 allele and 9 individuals (17%) with the "Yes" ApoE4 allele. These findings suggest that while the ApoE4 allele is more common in both clusters, its prevalence differs slightly between the two groups. The clusters may represent different risk profiles for dementia, with further analysis needed to better understand the clinical significance of these subgroups.

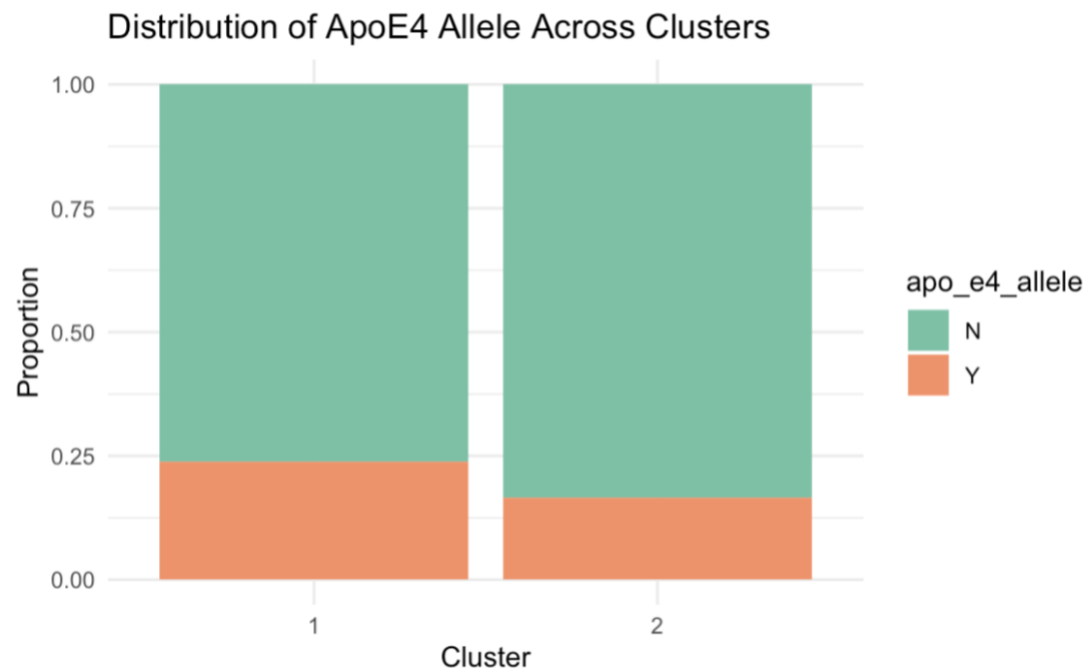


Figure 4.

Discussion

The combination of logistic regression, random forests, and clustering provided a comprehensive approach to understanding dementia risk in the study cohort. The random forest model outperformed the logistic regression model in terms of accuracy, sensitivity, and specificity, meaning greater classification capabilities. The logistic regression model provided insights into the relationship between predictors and the outcome, particularly highlighting the role of the ApoE4 allele in dementia risk.

The k-means clustering analysis revealed two distinct subgroups within the data, differentiated by the presence of the ApoE4 allele. These subgroups represent individuals with varying levels of risk for developing dementia, though further clinical validation is needed to understand the implications of these findings fully.

However, several limitations should be considered when interpreting the results. The sample size of 100 observations (after preprocessing) is relatively small, limiting the generalizability of the findings. A larger dataset would provide more robust estimates and improve the power of the models. Due to the small size, there is a higher risk of overfitting, and the models might not generalize well to other populations or settings. Additionally, the small sample size might also have impacted certain model parameters, particularly in the random forest analysis, where out-of-bag error rates were relatively high (44.29%). Another limitation is that the interpretability of the clusters is limited without additional clinical validation. The two clusters identified were distinguished primarily by the presence of the ApoE4 allele, but their clinical significance and how they relate to dementia risk need further investigation. Alternative clustering methods, such as hierarchical clustering could be explored in future studies to potentially uncover more nuanced subgroups.

Conclusion

Overall, this study demonstrates the potential of machine learning techniques to predict dementia risk and identify subgroups with different risk profiles based on clinical features. The random forest model performed well in classifying dementia cases, while the clustering analysis provided insight into potential subgroups at varying risk levels. Future work should focus on refining the clustering approach and exploring other advanced machine learning techniques to improve predictive accuracy. Additionally, the integration of genetic and neuroimaging data could further support our understanding of the factors contributing to dementia in TBI-exposed populations.

1. World Health Organization. (n.d.). *Dementia*. World Health Organization. <https://www.who.int/news-room/fact-sheets/detail/dementia>
2. Zhou, Y., Xu, J., & Liu, X. (2020). *The pathophysiology and clinical treatment of traumatic brain injury*. *Frontiers in Neurology*, 11, 573. <https://doi.org/10.3389/fneur.2020.00573>
3. Aging Brain Map. (n.d.). Allen Institute for Brain Science. <https://aging.brain-map.org/overview/home>